WORKSHEET

The quotient rule

1 Differentiate the following using the quotient rule.

a $\frac{1}{x}$

b $\frac{-3}{x}$

c $\frac{7}{x^2}$

d $\frac{5}{x-2}$

e $\frac{x+11}{7}$

f $\frac{x-1}{x}$

g $\frac{x}{x+1}$

h $\frac{x^2}{x+9}$

2 Differentiate the following functions.

a $\frac{2x}{5x-1}$

b $\frac{7x+3}{8x-1}$

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c $\dfrac{3x+9}{4x^2+5}$

d $\dfrac{7-2x}{x^2+3x-4}$

e $\dfrac{5x^2-7x+1}{x^3}$

f $\dfrac{2x^2-x+4}{x^2+3x-5}$

g $\dfrac{6x^4-2x^3}{2x-x^5}$

h $\dfrac{4x^2-7}{3x^2-2x+1}$

i $\dfrac{5x^2+x+8}{2x^2-7x-3}$

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Answers

1 **a** $\frac{-1}{x^2}$

b $\frac{3}{x^2}$

c $\frac{-14}{x^3}$

d $\frac{-5}{(x-2)^2}$

e $\frac{1}{7}$

f $\frac{1}{x^2}$

g $\frac{1}{(x+1)^2}$

h $\frac{x^2+18x}{(x+9)^2}$

2 **a** $\frac{-2}{(5x-1)^2}$

b $\frac{-31}{(8x-1)^2}$

c $\frac{-12x^2-72x+15}{(4x^2+5)^2}$

d $\frac{2x^2-14x-13}{(x^2+3x-4)^2}$

e $\frac{-5x^4+14x^3-3x^2}{x^6}$

f $\frac{7x^2-28x-7}{(x^2+3x-5)^2}$

g $\frac{6x^6-4x^5+36x^3-8x}{(2-x^4)^2}$

h $\frac{-8x^2+50x-14}{(3x^2-2x+1)^2}$

i $\frac{-37x^2-62x+53}{(2x^2-7x-3)^2}$

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